
3AL Test 2 2022

Time: 90 minutes

Available marks: 80

Maximum marks: 75

Instructions

1. Answer all questions in the lined answer booklets.
 2. Start each question on a new page.
 3. Make sure each answer is clearly labelled.
 4. Show all work and reasoning.
 5. This test consists of 4 questions. Make sure your question paper is complete.
 6. If you've read the instructions write your favourite Greek letter on the front of your booklet :) Good luck! Extra mark if you do
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1. (a) State the definition of a group action.
(b) State the definition of a stabiliser.
(c) Let $\varphi : G_1 \rightarrow G_2$ be a group homomorphism and let $*_2 : G_2 \times X \rightarrow X$ be a group action by G_2 on X .
(i) Prove that the map $*_1 : G_1 \times X \rightarrow X$ defined by $g *_1 x = \varphi(g) *_2 x$ is a group action.
(ii) Let $S_1(x)$ denote the stabiliser of $x \in X$ under the group action $*_1$. Show that $\ker \varphi \subseteq S_1(x)$ for all $x \in X$.

[5 + 5 + 5 + 5 = 20 marks]

NOTE: Be careful with notation in (c). Do not confuse $*_1$ with $*_2$, etc.

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2. (a) State the Class Equation.
- (b) State the definition of a p -group and give an example.
- (c) Let G be a p -group. Prove that $Z(G)$ is nontrivial.

[5 + 5 + 10 = 20 marks]

3. (a) State the Second Sylow Theorem.
- (b) Let G be a group of order $p^n m$ where $n \geq 1$ and p is a prime that does not divide m . Let H denote the intersection of all Sylow p -subgroups of G .
- (i) Show that if P is a Sylow p -subgroup of G and $g \in G$, then gPg^{-1} is a Sylow p -subgroup of G .
- (ii) Prove that $|H|$ is a power of p .
- (iii) Prove that H is normal in G .

[5 + 5 + 5 + 10 = 25 marks]

4. (a) State the Third Sylow Theorem.
- (b) Prove that a group of order 20 has a normal subgroup.

[5 + 10 = 15 marks]